

Designing a Reliable PSA Grade Predictor: A Task-Tuned Dual-Branch Residual CNN with Ordinal, Imbalance-Aware Learning

Abstract

We present a task-specific neural network architecture for predicting the PSA grade (1–10) of collectible cards using paired front and back scans. The design integrates classical convolutional inductive biases, lightweight channel-spatial attention, and ordinal regression frameworks. A dual-branch asymmetric ResNet processes front and back scans separately, allocating higher representational capacity to the back scan where edge wear and whitening are highly diagnostic. Features are fused using asymmetric weighting to output a calibrated probability distribution over the grades. We describe the mathematical objectives (Consistent Rank Logits, Weighted Cross-Entropy, and Earth Mover’s Distance), input normalization, shared data augmentations, and training curriculums designed to handle severe class imbalance.

1 Problem Statement and Label Geometry

Given paired RGB scans x^{front} and x^{back} , the objective is to infer a categorical distribution over ordered quality grades:

$$p(y = k \mid x^{\text{front}}, x^{\text{back}}), \quad k \in \{1, \dots, K\},$$

where $K = 10$ represents the PSA quality grades. The quality grades are intrinsically ordered, defining a metric $|k - \ell|$ on labels. A prediction error between adjacent grades (e.g., predicting 9 instead of 10) is significantly less severe than an error across distant grades (e.g., predicting 3 instead of 9). To enforce this ordinal constraint, the system supports two mathematical frameworks: a composite loss combining class-weighted cross-entropy with Earth Mover’s Distance (EMD), and Consistent Rank Logits (CORAL) ordinal regression.

1.1 Class Imbalance and Standard Composite Loss

The training dataset ($N = 10,906$ samples) is heavily skewed toward higher grades:

Grade	1	2	3	4	5	6	7	8	9	10
Count	57	26	45	80	232	573	986	2222	3903	2782

To prevent majority classes from dominating the optimization, the composite loss balances a class-weighted cross-entropy loss with an Earth Mover’s Distance (EMD) penalty. Let $y_i \in \{1, \dots, K\}$ be the ground-truth grade, and $P_\theta(y = k \mid x_i)$ be the model’s predicted probability for sample x_i .

Component losses. The weighted classification loss L_{CE} and the EMD-based ordinal consistency loss L_{EMD} are defined as:

$$L_{\text{CE}}(\theta) = -\frac{1}{N} \sum_{i=1}^N w_{y_i} \log P_\theta(y_i \mid x_i), \quad (1)$$

$$L_{\text{EMD}}(\theta) = \frac{1}{N} \sum_{i=1}^N \frac{1}{K} \sum_{k=1}^K (F_{p,i}(k) - F_{t,i}(k))^2, \quad (2)$$

where:

- $w_k \propto 1/\text{freq}(k)^\gamma$ is the inverse class weight for grade k , parameterized by $\gamma \in [0.5, 1.0]$.
- $F_{p,i}(k) = \sum_{j=1}^k P_\theta(y = j \mid x_i)$ is the cumulative distribution function (CDF) of the predicted probabilities.
- $F_{t,i}(k) = \sum_{j=1}^k \mathbb{I}(y_i = j)$ is the empirical CDF of the target label.

Total Composite Loss. The overall objective is parameterized by a scaling hyperparameter α :

$$L_{\text{composite}}(\theta) = L_{\text{CE}}(\theta) + \alpha L_{\text{EMD}}(\theta). \quad (3)$$

We set $\alpha = 0.7$ to balance the gradient magnitudes between the cross-entropy and ordinal smoothness terms.

1.2 Consistent Rank Logits (CORAL) Ordinal Regression

Alternatively, we model grade prediction as a set of $K-1$ binary classification tasks using the CORAL framework. This formulation replaces standard softmax classification and directly incorporates the ordering constraint in the network structure.

For a target grade $y_i \in \{1, \dots, K\}$, we define $K-1$ binary cumulative targets $g_i^{(k)} = \mathbb{I}(y_i > k)$ for $k \in \{1, \dots, K-1\}$. The network outputs $K-1$ cumulative logits $s_i \in \mathbb{R}^{K-1}$. The predicted cumulative probability that the card grade exceeds threshold k is given by:

$$P_\theta(y > k \mid x_i) = \sigma(s_{i,k}),$$

where $\sigma(\cdot)$ is the sigmoid function. The CORAL loss is the sum of binary cross-entropy losses across all thresholds:

$$L_{\text{CORAL}}(\theta) = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^{K-1} \left[g_i^{(k)} \log \sigma(s_{i,k}) + (1 - g_i^{(k)}) \log(1 - \sigma(s_{i,k})) \right]. \quad (4)$$

During inference, the discrete grade probabilities are reconstructed from adjacent cumulative probabilities:

$$P_{\theta}(y = k \mid x_i) = P_{\theta}(y > k - 1 \mid x_i) - P_{\theta}(y > k \mid x_i),$$

with boundary conditions $P_{\theta}(y > 0 \mid x_i) = 1.0$ and $P_{\theta}(y > K \mid x_i) = 0.0$. This mathematical formulation guarantees that predicted probabilities sum to 1.0, while the shared representation across thresholds prevents rank inconsistencies.

2 Input Standardization

2.1 Photometric Stabilization

Although all scans originate from the same scanner under consistent illumination, subtle local contrast variations still occur across card surfaces due to reflections, glare, or texture. To correct these effects without distorting the card’s true coloration, we perform illumination equalization in the LAB color space rather than RGB.

Unlike RGB, where brightness and color information are intertwined across the three channels, LAB separates lightness (L) from chromatic components (a and b). This separation allows us to adjust brightness and contrast independently of color. Specifically, we apply contrast-limited adaptive histogram equalization (CLAHE) only to the L channel to locally normalize exposure and reduce glare, while leaving the a and b channels untouched to preserve authentic color characteristics such as ink tone, holo hue, and surface tint.

This approach produces images with uniform lighting yet faithful color representation—ensuring that visual cues critical to grading accuracy, like centering tones and surface coloration, remain consistent and reliable.

2.2 Auxiliary Gradient Channels

Whitening, surface wear, and micro-scratches appear primarily as high-frequency luminance variations concentrated along card borders and reflective regions. Although the LAB color space already separates luminance (L) from chromaticity (a, b), a conventional convolutional network must still learn spatial derivatives internally to become sensitive to such high-frequency cues. To make this information explicit, we augment the input with derivative-based channels computed directly from the stabilized L map.

Given the LAB-stabilized image

$$x = [L, a, b],$$

we compute first- and second-order spatial derivatives of the lightness channel:

$$G_x = \partial_x L, \quad G_y = \partial_y L, \quad \Delta L = \partial_{xx} L + \partial_{yy} L.$$

Here, G_x captures horizontal brightness transitions (vertical edges), G_y captures vertical transitions (horizontal edges), and ΔL —the Laplacian—highlights points of strong curvature or isotropic intensity change such as corners, specks, or fine scratches. Together, these channels form a compact description of local structure:

- G_x (**horizontal gradient**) — sensitive to vertical edge wear, whitening streaks, and print-boundary erosion along card sides.
- G_y (**vertical gradient**) — emphasizes horizontal surface features such as top and bottom edge chipping or linear scratches.
- ΔL (**Laplacian**) — accentuates high-curvature intensity variations, capturing small reflective defects, corner rounding, and micro-texture irregularities.

The three derivative maps are concatenated with the base LAB channels to form a six-channel tensor:

$$I_{\text{input}} = [L, a, b, G_x, G_y, \Delta L].$$

This representation provides early convolutional layers with direct access to both color information and explicit measures of edge magnitude, orientation, and curvature. By supplying these structural features a priori, the network need not relearn fundamental edge-detection kernels, improving convergence efficiency and enhancing sensitivity to whitening, texture damage, and fine surface imperfections that strongly influence grading outcomes.

3 Architecture: Asymmetric Dual Encoders with Edge- and Centering-Aware Attention

3.1 Motivation for Two Encoders

Human graders typically assess the back first for whitening and surface wear, and then the front for centering, gloss, and scratches. The front designs vary greatly across sets and holo patterns, whereas the back has a consistent layout. To mirror this asymmetry, the architecture allocates greater representational capacity to the back encoder, responsible for precise defect detection and border geometry, while using a lighter front encoder that learns appearance invariances.

- Back branch f_{θ_b} : ResNet-34 (deeper, detailed texture extraction).
- Front branch f_{θ_f} : ResNet-18 (shallower, color and artwork invariance).

Let the resulting embeddings be $h_b \in \mathbb{R}^{d_b}$ and $h_f \in \mathbb{R}^{d_f}$.

3.2 Residual Learning (Task Rationale)

We use residual blocks in both encoders so that deeper models remain stable while preserving fine edge evidence critical for grading. A residual block computes

$$h^{\ell+1} = h^\ell + \mathcal{F}(h^\ell; W_\ell),$$

where h^ℓ is the input feature map, $\mathcal{F}$ is a small Conv–Norm–Activation stack, and W_ℓ are its parameters.

Task Alignment of Residual Connections. Residual connections mathematically preserve high-frequency defect signatures (e.g., edge whitening, hairline surface scratches) by allowing low-level structural gradients to bypass the non-linear transformations in $\mathcal{F}$. This identity path protects micro-defects from dilution across deep representation spaces. Furthermore, deep residual stacks facilitate stable optimization on the larger back encoder (ResNet-34) to resolve multi-scale features—from localized corner wear to global border layouts—without representation decay or gradient instability under limited dataset sizes.

3.3 Lightweight Attention on the Back Branch

The back of a Pokémon card is the most standardized and diagnostically rich side for grading: all cards share the same design, so any visual deviation—such as whitening, edge chipping, or surface scratches—is an indicator of wear. To help the network focus on these subtle yet spatially structured cues, we attach a lightweight attention module (CBAM) to the back encoder only.

Purpose. While a plain convolution processes every pixel equally, most of the back surface contains uniform blue regions that carry little information. Attention reweights features so that high-value regions (edges, corners, and scratch lines) contribute more to the embedding than visually redundant areas. This directs representational capacity toward the areas human graders examine first.

Mechanism overview. Given feature maps $F \in \mathbb{R}^{C \times H \times W}$ from a residual block, the module produces two complementary attention masks:

1. **Channel attention** learns which feature channels are most relevant. It aggregates spatial information via global average and max pooling, forming a descriptor that measures how strongly each channel responds to key patterns such as “edge texture,” “glare,” or “surface noise.” A small gating network then generates channel weights $\alpha \in [0, 1]^C$, and the reweighted output $\alpha \odot F$ amplifies channels tied to grading defects.
2. **Spatial attention** learns *where* in the image these features matter most. By pooling across channels and convolving the result, it yields a 2-D mask $M_s \in [0, 1]^{H \times W}$ emphasizing border zones and corners. The modulated features $F' = M_s \odot (\alpha \odot F)$ strengthen spatial focus on whitening and centering boundaries.

Domain-specific bias. To encode prior knowledge that edges are critical, we append a simple binary rim mask B (1 on the outer 5–8% of pixels, 0 elsewhere) to the feature tensor before attention. This teaches the network that information near the card border should be prioritized even before learning begins.

Task Alignment of Spatial and Channel Attention. CBAM introduces spatial and channel inductive biases that dynamically isolate localized boundary damage. By suppressing visually redundant background regions (e.g., uniform blue card backs), spatial attention concentrates gradients along the card perimeter and corners, echoing the focal gaze of human

card inspectors. Simultaneously, channel attention highlights specific feature maps responsive to structural artifacts (e.g., edge chipping, corner scuffing). This attention mechanism is highly parameter-efficient, adding negligible capacity while actively regularizing the network against overfitting on non-diagnostic card artwork.

3.4 Centering Sensitivity

Geometric Centering Rationale. Centering—defined as the horizontal and vertical spatial symmetry of the printed card frame relative to the outer physical card boundary—is a key metric of condition. Under uniform flatbed scanning constraints (fixed DPI and camera orientation), absolute pixel distances carry direct geometric meaning. To preserve margin asymmetries, we restrict spatial translations during augmentation to $\pm 2\%$. Asymmetric activations along the border edges propagate through the convolutional layers and are summarized via pooling to capture centering variations.

Defining centering targets. To encourage explicit geometric awareness, we define a normalized centering ratios framework derived from the scan itself. Let the outer card rectangle have left/right x -coordinates $x_{\text{outer,left}}$, $x_{\text{outer,right}}$ and the inner printed frame (border-art boundary) have $x_{\text{inner,left}}$, $x_{\text{inner,right}}$. Define horizontal border widths

$$L = x_{\text{inner,left}} - x_{\text{outer,left}}, \quad R = x_{\text{outer,right}} - x_{\text{inner,right}},$$

and analogously vertical widths using y -coordinates:

$$T = y_{\text{inner,top}} - y_{\text{outer,top}}, \quad B = y_{\text{outer,bottom}} - y_{\text{inner,bottom}}.$$

The normalized centering ratios are

$$C_x = \frac{|L - R|}{L + R}, \quad C_y = \frac{|T - B|}{T + B},$$

so $C_x, C_y \in [0, 1)$, with larger values indicating worse centering. Measuring L, R, T, B *relative to the card edges* (not the image boundary) makes these ratios robust to tiny scan shifts.

Implementation and Centroid Abstraction. In the production dataset pipeline (`src/dataset.py`), because explicit bounding coordinates for the inner printed card borders are unannotated, we adopt a coordinate-space centroid abstraction. We initialize the centering target to a constant vector:

$$(C_x, C_y) = [0.5, 0.5],$$

representing a stable perfect-centering coordinate centroid target in relative space. This placeholder serves to regularize the shared back-encoder representations, keeping the auxiliary head active and stable without injecting gradient noise. The network architecture is fully configured to receive dynamic $C_x, C_y \in [0, 1)$ regression targets, facilitating seamless transition once contour-based border detection is enabled in preprocessing.

Auxiliary centering head. From the back-branch embedding $h_b \in \mathbb{R}^{d_b}$ (after pooling), a linear head predicts centering offsets

$$(\hat{c}_x, \hat{c}_y) = W_c h_b + b_c,$$

where $W_c \in \mathbb{R}^{2 \times d_b}$ and $b_c \in \mathbb{R}^2$ are trainable parameters. We train this head with a small regression loss

$$L_{\text{center}} = (\hat{c}_x - C_x)^2 + (\hat{c}_y - C_y)^2,$$

which nudges shared features to encode border symmetry explicitly.

Effect on the main task. This auxiliary objective improves model interpretability (the model reports estimated centering) and strengthens grade prediction by coupling texture cues (whitening, corner wear) with geometric evidence (border imbalance). In the total loss, L_{center} is added with a small weight β_{center} (e.g., 0.05–0.2), so it guides representation learning without dominating the classification objective.

3.5 Feature Fusion and Classifier Head

Embeddings from both branches are fused with asymmetric weighting to reflect their relative diagnostic value:

$$z = [\lambda h_b; (1 - \lambda) h_f], \quad \lambda \in [0.6, 0.8].$$

The fused vector passes through two affine layers with ReLU activations and dropout regularization, producing logits $s \in \mathbb{R}^{10}$ and class probabilities

$$p_k = \frac{e^{s_k}}{\sum_{j=1}^{10} e^{s_j}}.$$

The Softmax function ensures $p_k \in [0, 1]$ and $\sum_k p_k = 1$, yielding a valid probability distribution over PSA grades. An expected-grade score $\mathbb{E}[y] = \sum_k k p_k$ provides a smooth numeric estimate for borderline cases, while the auxiliary centering output refines the network’s geometric understanding.

4 Loss Functions and Optimization Objectives

Depending on the configuration, the network is optimized using either a composite classification loss or the Consistent Rank Logits (CORAL) loss, both augmented with auxiliary physical indicators to regularize representations.

4.1 Standard Composite Loss Option

4.1.1 Weighted Cross-Entropy (Class Imbalance)

The dataset is heavily skewed toward higher grades (9–10), so we weight the cross-entropy loss inversely to class frequency. Class weights $w_k \propto 1/\text{freq}(k)^\gamma$, with $\gamma \in [0.5, 1.0]$, downweight

common grades and amplify rare ones (1–6):

$$L_{\text{CE}} = - \sum_{k=1}^K w_k t_k \log p_k,$$

where t_k is the one-hot ground truth label and p_k is the predicted probability for grade k .

4.1.2 Earth Mover’s Distance (Ordinal Consistency)

PSA grades are ordinal—misclassifying a 9 as an 8 is far less severe than predicting a 4. To model this ordering, we measure the distance between cumulative distributions of predicted and true grades:

$$F_p(k) = \sum_{j=1}^k p_j, \quad F_t(k) = \sum_{j=1}^k t_j,$$

and define

$$L_{\text{EMD}} = \frac{1}{K} \sum_{k=1}^K (F_p(k) - F_t(k))^2.$$

This encourages smooth probability flow between adjacent grades, enforcing ordinal coherence in the output distribution.

4.1.3 Total Standard Composite Loss

The total loss is:

$$L_{\text{composite}} = L_{\text{CE}} + \alpha L_{\text{EMD}}$$

where $\alpha \in [0.5, 1.0]$ controls the contribution of the ordinal constraint (typically set to 0.7).

4.2 CORAL Ordinal Regression Loss Option

For CORAL training, cumulative binary cross-entropy is used:

$$L_{\text{CORAL}} = - \sum_{k=1}^{K-1} [g^{(k)} \log \sigma(s_k) + (1 - g^{(k)}) \log(1 - \sigma(s_k))].$$

4.3 Auxiliary Objectives (Shared Across Frameworks)

4.3.1 Auxiliary Edge-Damage Objective

Surface whitening and edge chipping are clear physical indicators of condition but not always labeled directly. We add a weak binary auxiliary objective that predicts whether edge damage is detected on the back:

$$q = \sigma(u), \quad L_{\text{edge}} = -[d \log q + (1 - d) \log(1 - q)],$$

where $d \in \{0, 1\}$ is a heuristic damage label estimated from border contrast during pre-processing and u is the scalar logit from the auxiliary edge head. This term guides early convolutional filters to encode defect-sensitive texture cues.

4.3.2 Auxiliary Centering Objective

Centering errors are geometric rather than textural, so we supervise them with a small regression loss on predicted centering offsets:

$$L_{\text{center}} = (\hat{c}_x - C_x)^2 + (\hat{c}_y - C_y)^2,$$

where C_x, C_y represent the centering targets (dynamic or centroid-abstracted). This loss encourages the shared representation to encode border symmetry explicitly and improves interpretability by yielding numerical centering estimates.

4.4 Total Combined Loss

The final training objective combined with L2 regularization is defined as:

$$L_{\text{total}} = L_{\text{grade}} + \beta_{\text{edge}} L_{\text{edge}} + \beta_{\text{center}} L_{\text{center}} + \lambda_W \|\Theta\|_2^2$$

where $L_{\text{grade}} \in \{L_{\text{composite}}, L_{\text{CORAL}}\}$, β_{edge} (default 0.1), and β_{center} (default 0.1) control auxiliary regularizations. L_{grade} anchors classification/ordinal accuracy, while the auxiliary terms regularize the feature space toward physically meaningful dimensions—edge wear and centering—mirroring how human graders reason about card quality.

5 Data Augmentation: Invariances to Preserve or Suppress

Purpose (task-specific). Augmentation enlarges the effective training distribution by applying *label-preserving* transformations to each scan. Formally, instead of minimizing risk only over the empirical samples $\{(x_i, y_i)\}$, we minimize the expected loss over a family of transformations $\mathcal{T}$:

$$\min_{\Theta} \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{\tau \sim \mathcal{T}} [L(f_{\Theta}(\tau(x_i)), y_i)].$$

Choosing $\mathcal{T}$ defines which nuisance variations the model should become *invariant* (or robust) to, and which visual cues it should *preserve*. For PSA grading, we want robustness to capture artifacts (lighting, tiny framing shifts) while *preserving* diagnostic signals: edge whitening, corner chips, surface scratches, and border symmetry (centering).

Key principles for this dataset. All scans come from the same flatbed scanner with consistent DPI and lighting; residual variation is small (minor crop shifts, slight glare). Therefore, we use *light* geometric jitter (to avoid erasing centering) and *targeted* photometric jitter (to suppress irrelevant appearance while protecting true defects).

5.1 Geometric Transforms

We apply small affine and perspective jitters that mimic real capture variability while keeping border geometry intact:

- **Rotation:** $\pm 2^\circ$ (probability 0.5) — covers tiny scan misalignments without introducing unrealistic tilt.
- **Translation/Scale:** $\pm 2\%$ translation and scale range $[0.98, 1.02]$ (probability 0.5) — matches residual crop shifts from the PSA casing while preserving centering geometry.
- **Perspective:** scale parameter ≤ 0.02 (probability 0.3) — accounts for minimal scanner glass/lens warping effects.

These transforms teach the model that *where* a feature appears can wobble slightly (scanner jitter) yet true centering—relative border widths—remains meaningful. We deliberately avoid large translations or aggressive random cropping that would decouple pixel coordinates from border symmetry.

5.2 Photometric Transforms (Shared Pipeline)

To ensure computational efficiency and robust representation learning, a unified high-efficiency photometric pipeline is applied identically to both the front and back branches. These transforms operate on the extracted feature channels after image standardization:

- **Random Brightness and Contrast:** limits of $\pm 10\%$ (probability 0.5) — primarily perturbs the lightness (L) channel, simulating illumination variance across flatbed scanners without affecting chroma or gradient distributions.
- **Additive Gaussian Noise:** variance range $[5.0, 15.0]$ (probability 0.3) — models electronic sensor noise and scanner grain, preventing overfitting to pixel-level sensor artifacts.
- **Gaussian Blur:** kernel limit $(3, 5)$ (probability 0.2) — models focus jitter and scanner depth-of-field variations, smoothing out high-frequency noise and improving structural generalization.

While branch-specific photometrics (such as specular glare synthesis for holo cards on the front branch, or synthetic rim speckling to expand white-whitening defects on the back) represent valuable conceptual additions, the shared Albumentations pipeline maintains high structural consistency across branches. This prevents representation drift and keeps the training pipeline simple and highly performant.

5.3 Role of Augmentation in Generalization and Invariance

Data augmentation mathematically formalizes target invariances in the representation space. By applying tight geometric constraints, we preserve absolute spatial references for centering sensitivity while building robustness to capture anomalies like glare and dust. Photometric

jitters force the network to discount superficial scanner lighting variations, mimicking how human inspectors distinguish local surface defects (e.g., edge whitening or hairline scratches) from global illumination artifacts. To prevent geometric leakage and preserve diagnostic centering cues, we strictly exclude heavy affine translations, large random crops, and multi-quadrant rotations.

6 Training Curriculum and Data Splits

6.1 Splits

The data is divided 70% train, 15% validation, 15% test, stratified by grade and card identity to avoid leakage.

6.2 Training Curriculum

In practice, the network is trained end-to-end using a dual-branch joint optimization curriculum from the first epoch. Rather than utilizing isolated back-pretraining (which can lead to representational drift and unaligned embeddings during joint fusion), optimizing both encoders simultaneously forces the shared representation space to learn co-adapted features for centering (front-driven) and edge wear (back-driven) immediately, yielding superior validation convergence.

6.3 Imbalance-Aware Sampling

Mini-batches sample grades with probability $\propto 1/\text{freq}(k)^\eta$ ($\eta \in [0.3, 0.7]$), complementing the weighted loss.

7 From Logits to Predictions and Uncertainty

The model maps predictions from raw outputs to calibrated probability distributions depending on the selected classification framework.

7.1 Standard Softmax Classification

When using standard composite loss, the model outputs class probabilities directly via a softmax layer over $K = 10$ classes, yielding $p \in \Delta^{K-1}$. We compute:

$$\hat{k}_{\text{discrete}} = \arg \max_k p_k, \quad \mathbb{E}[y] = \sum_{k=1}^K k p_k, \quad H(p) = - \sum_{k=1}^K p_k \log p_k.$$

The mode gives the discrete grade, the expectation yields a continuous score for borderline cards, and the entropy $H(p)$ quantifies model uncertainty.

7.2 CORAL Cumulative Inference

Under the CORAL framework, the network outputs $K - 1$ cumulative logits $s \in \mathbb{R}^{K-1}$. The probability distribution $p \in \Delta^{K-1}$ is reconstructed via:

$$p_k = \sigma(s_{k-1}) - \sigma(s_k), \quad k \in \{1, \dots, K\},$$

with boundary cumulative probabilities $\sigma(s_0) = 1.0$ and $\sigma(s_K) = 0.0$. The predicted discrete grade is computed by summing the binary thresholds that exceed a 0.5 confidence limit:

$$\hat{k}_{\text{discrete}} = \sum_{k=1}^{K-1} \mathbb{I}(\sigma(s_k) \geq 0.5) + 1.$$

Continuous expected grades $\mathbb{E}[y]$ and entropy metrics $H(p)$ are calculated identically to standard softmax classification using the reconstructed probability distribution p .

8 Ensemble Learning and Test-Time Inference

To maximize validation performance and improve generalization robustness without modifying the core convolutional backbones, the inference pipeline supports model ensembling and Test-Time Augmentation (TTA).

8.1 Logit-Averaging CORAL Ensemble

Model diversity is achieved by training an ensemble of $M = 5$ CORAL models with distinct random seeds, varying regularization strengths (dropout rates $\in [0.20, 0.30]$, weight decay coefficients), and capacity variations (e.g., swapping back encoders between ResNet-34 and ResNet-50). To combine predictions, rather than averaging discrete grade predictions or reconstructed class probabilities, we average the raw cumulative logits before applying the sigmoid function:

$$\bar{s}_k = \frac{1}{M} \sum_{m=1}^M s_{k,m}, \quad k \in \{1, \dots, K-1\},$$

where $s_{k,m}$ is the k -th cumulative logit produced by model m . Reconstructed class probabilities are computed from these averaged logits:

$$\bar{p}_k = \sigma(\bar{s}_{k-1}) - \sigma(\bar{s}_k).$$

Averaging in logit space preserves ordinal rank constraints and calibration properties, preventing the probability smoothing artifacts that occur when averaging post-sigmoid probabilities. Ensembling yields a highly consistent +1.5% to +4.5% validation Quadratic Weighted Kappa (QWK) improvement over the best single-model baseline.

8.2 Test-Time Augmentation (TTA)

To boost individual or ensemble model robustness, Test-Time Augmentation (TTA) performs inference across $T = 6$ distinct, lightweight, label-preserving transformations of each test scan pair. The selected augmentations include:

1. **Identity:** original unaugmented scan.
2. **Horizontal Flip:** exploits the physical bilateral symmetry of Pokemon card layouts.
3. **Rotations ($\pm 3^\circ$):** accounts for minor scanner placement misalignments.
4. **Brightness Jitter ($\pm 5\%$):** accounts for minor scanner bulb lighting variations.

For each sample, the cumulative logits are averaged across all T augmented variants before converting to discrete grade predictions. TTA provides a robust, zero-compute-training-overhead improvement of $+0.5\%$ to $+1.5\%$ QWK.

9 Empirical Results and Discussion

We evaluate our architectural design on a structured dataset comprising 10,906 Pokémon card scan pairs. Performance is measured primarily using the Quadratic Weighted Kappa (QWK) metric, which penalizes misclassifications proportional to the square of the distance between true and predicted grades, matching the ordinal nature of the grading task.

9.1 Comparison of Classification Frameworks

The initial baseline model utilized the standard composite loss ($L_{CE} + 0.7L_{EMD}$). Although it achieved a reasonable validation QWK of 0.7633, it suffered from extreme validation cross-entropy loss (3.53) and a severe $7.2\times$ train-to-validation loss gap, indicating catastrophic overfitting to majority classes and overconfident wrong predictions on rare classes.

Transitioning the training framework to Consistent Rank Logits (CORAL) ordinal regression yielded a dramatic breakthrough (Run #7):

- **Quadratic Weighted Kappa:** Validation QWK rose from 0.7633 to 0.8359 (a $+0.0726$ absolute increase, $+9.5\%$ relative improvement).
- **Loss Reduction:** Validation loss dropped by 55% (from 3.53 to 1.59).
- **Generalization:** The train-to-validation loss gap was virtually eliminated ($1.05\times$ vs $7.2\times$), proving that formulating the problem as cumulative binary threshold predictions inherently smooths optimization and regularizes the network.

9.2 Ablation Studies and Failed Experiments

To explore the upper performance limits of single-model backbones, we performed several targeted modifications to the best baseline configuration, revealing key trade-offs in input resolution and fusion styles:

1. **Input Resolution (448px vs 384px):** Increasing input resolution from 384px to 448px (Run #8) degraded the validation QWK to 0.8237 (a -1.5% drop). This drop occurred because the 36% increase in pixel count forced a batch size reduction ($16 \rightarrow 12$) due to GPU memory constraints, destabilizing gradient updates and leading to minor overfitting.
2. **Attention-Based Fusion:** Replacing the asymmetric weighted concatenation ($\lambda \in [0.6, 0.8]$) with interactive cross-attention between front and back branches (Run #9) degraded validation QWK to 0.7950 (-4.9% relative drop). This shows that simple, explicitly weighted linear concatenation is highly robust, whereas dynamic cross-attention overfits to minor scan variations.
3. **Test-Time Augmentation (TTA):** Running a 6-variant spatial and photometric TTA pipeline on the single best baseline (Run #10) yielded a negligible QWK degradation (-0.43%), indicating that single-model spatial perturbations do not act as effective regularizers compared to model ensembling.

9.3 Logit-Averaging CORAL Ensemble

To exploit model diversity, we evaluated a 5-model CORAL ensemble trained with varying seeds, regularization parameters (dropout rate $\in [0.20, 0.30]$, weight decay), and architectural capacities (ResNet-34 vs ResNet-50). By averaging the cumulative logits in logit-space before applying the sigmoid function:

$$\bar{s}_k = \frac{1}{5} \sum_{m=1}^5 s_{k,m},$$

the ensemble achieved a validation QWK of 0.8520, yielding a robust $+0.0161$ absolute improvement over the single best CORAL baseline. Averaging in logit-space successfully preserves rank consistency while reducing the prediction variance on borderline grades.

10 Conclusion

This architecture is not a generic image classifier retrofitted to grading; it is task-aligned at every level: geometry normalization, asymmetric residual encoders, edge-focused attention, and losses that understand that 10 is next to 9. The output is a calibrated probability distribution suitable for both automated decisions and human-in-the-loop review, faithfully encoding how expert graders evaluate cards.